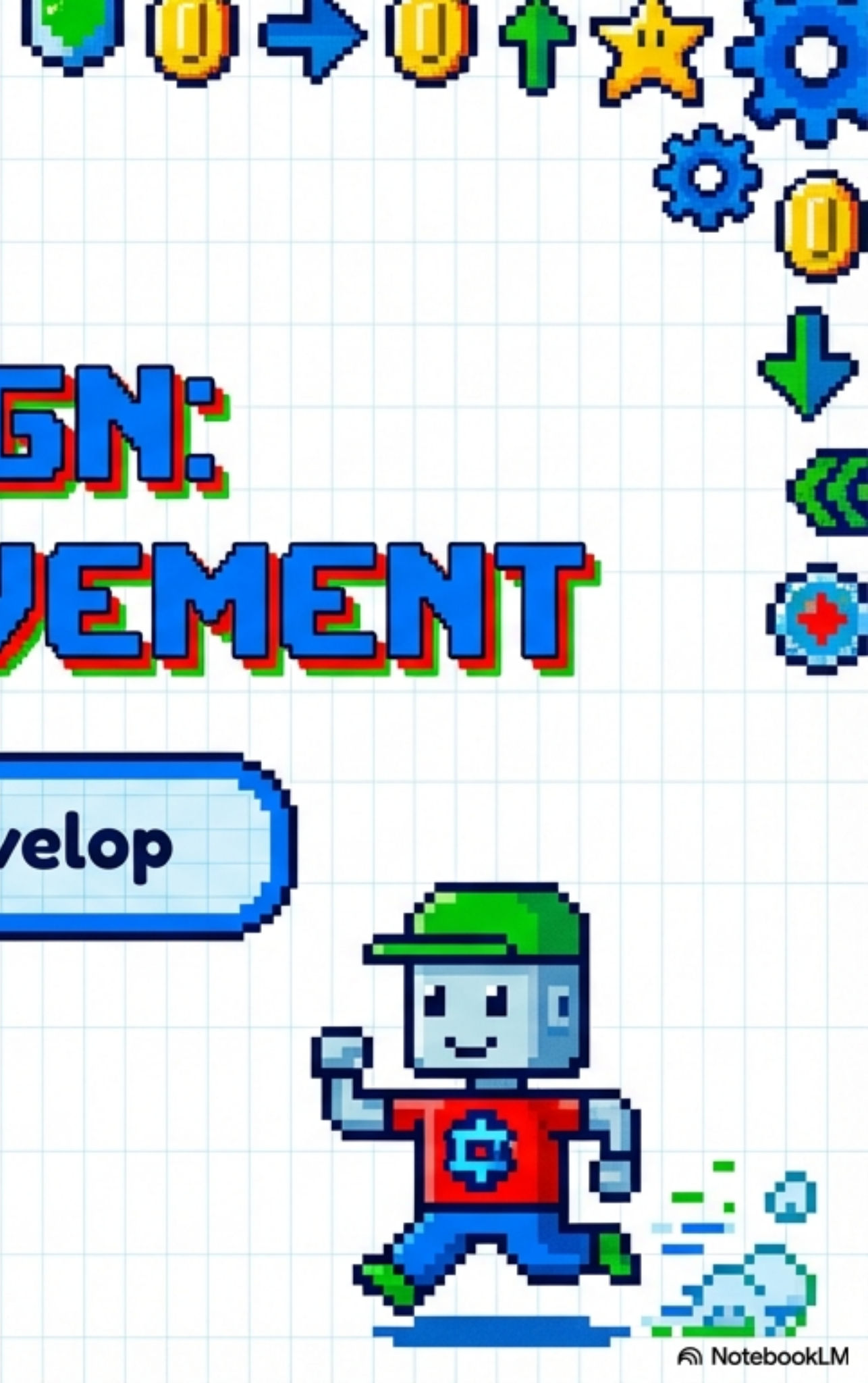




GAME DESIGN: MASTERING MOVEMENT

G5 - Lesson 7 | Using GDevelop



PRESS START



WOULD A GAME BE FUN IF NOTHING MOVED?

BORING



NO ACTION. NO CHALLENGE.
JUST PIXELS ON A SCREEN.



FUN



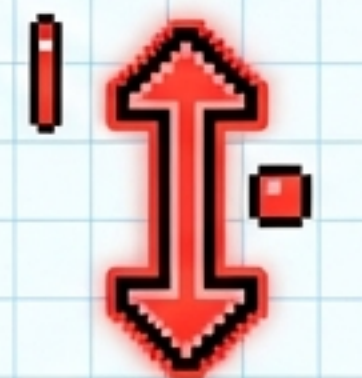
HIGH SCORE

ACTION! CHALLENGE! ADVENTURE!
PIXELS COME TO LIFE.

Mission Goal

Today's Mission: Recreate Vertical Movement
(Up & Down) like in the game Pong.

START



Vertical

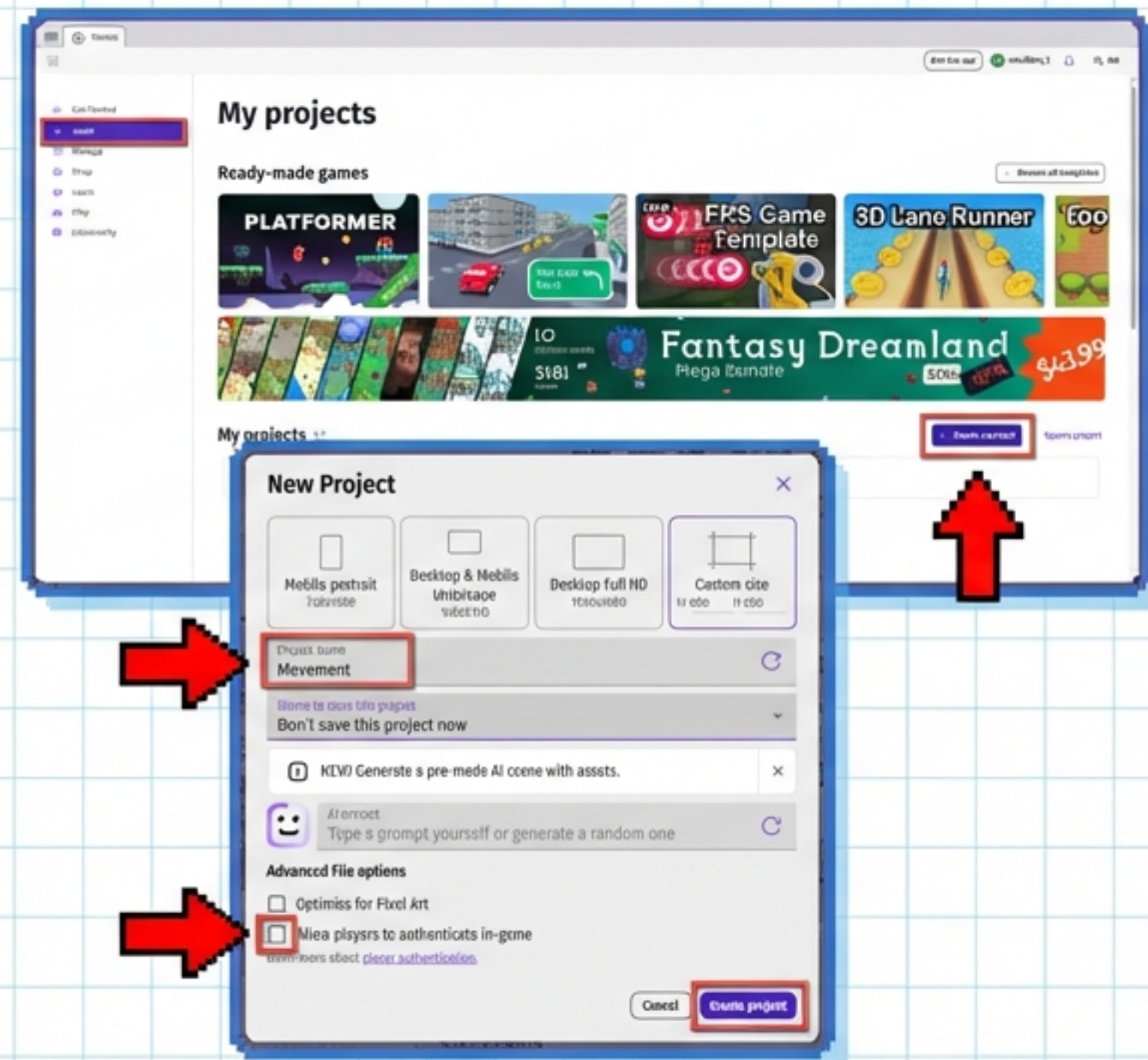


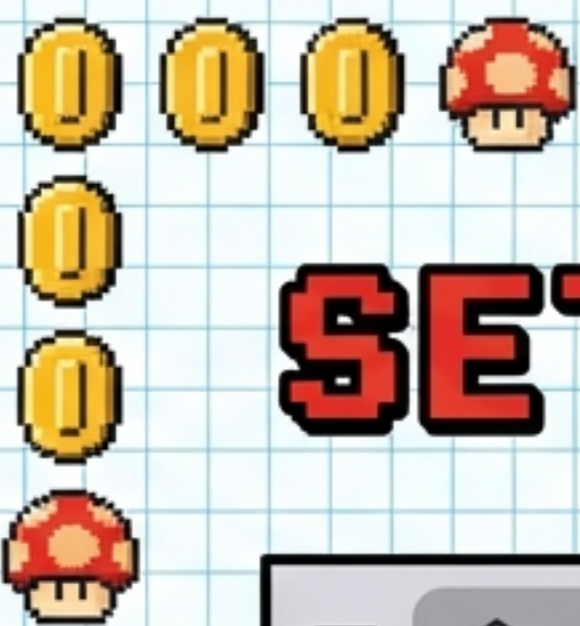
Horizontal

START

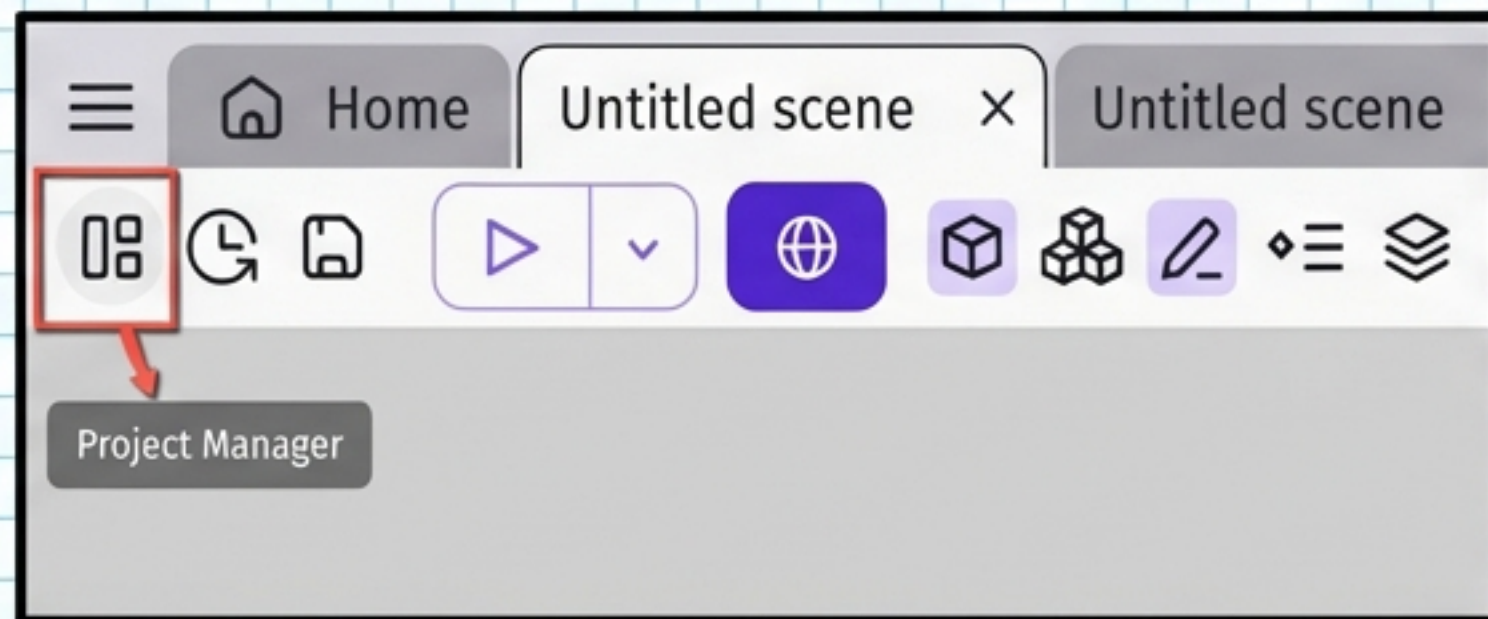
MISSION START: PROJECT SETUP

1. Select 'Create a project'.
2. Name your mission: 'Movement'.
3. **CRITICAL:** Uncheck 'Authenticate in-game'.
4. Click 'Create project'.





SETTING THE STAGE



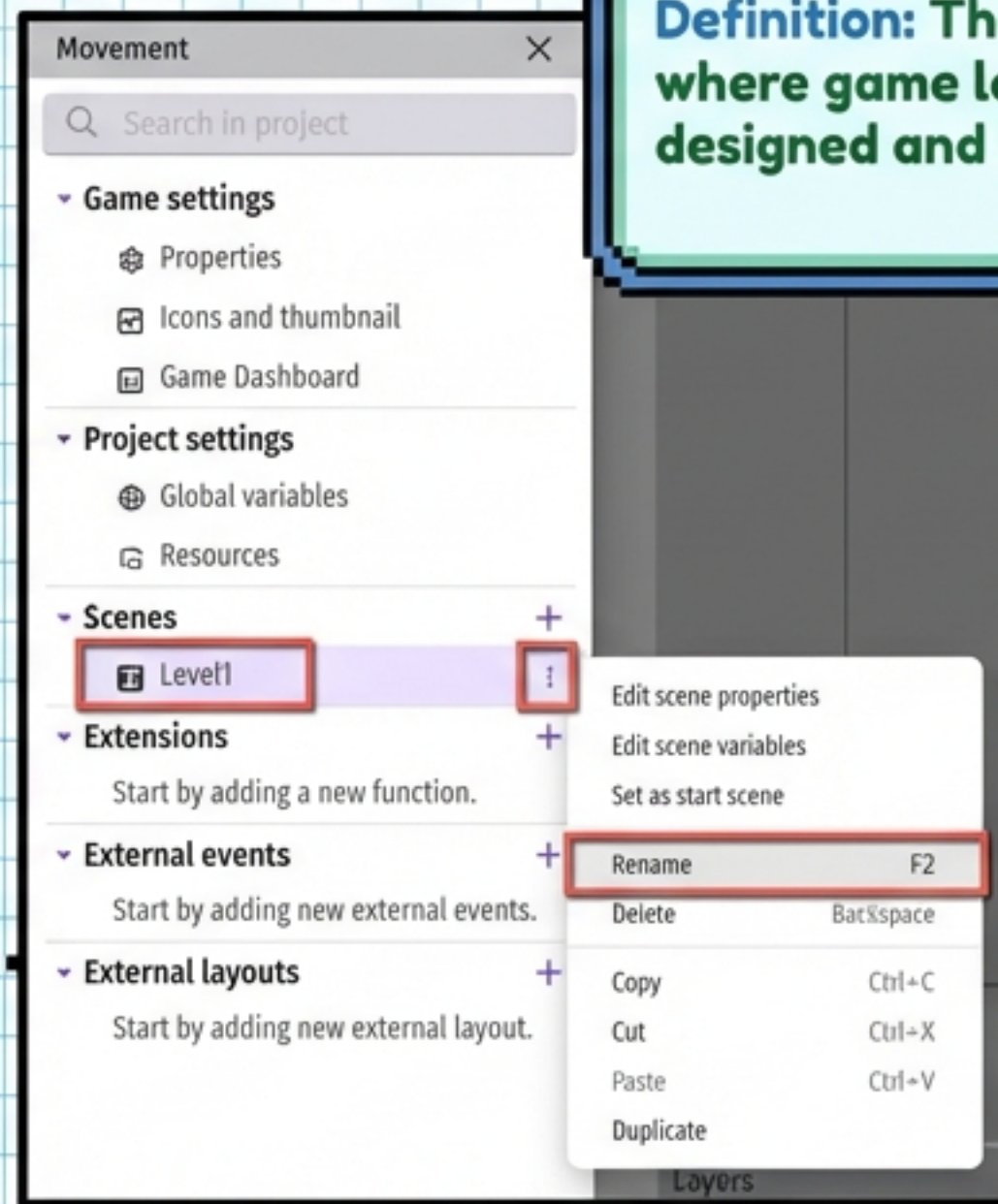
Action:

1. Open Project Manager.
2. Rename 'Untitled Scene' to 'Level1'.



Pro Tip:

Always name your scenes to stay organized!

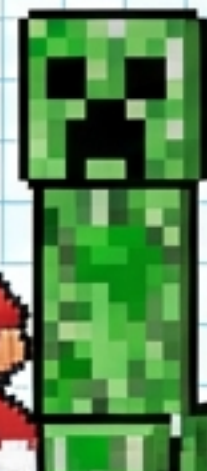
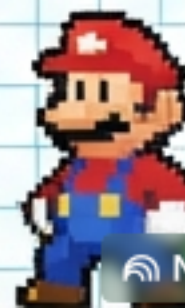


Tech Flash



ITEM ACQUIRED:
SCENE

Definition: The area
where game levels are
designed and built.





MEET THE ACTORS: OBJECTS & SPRITES



Tech Flash



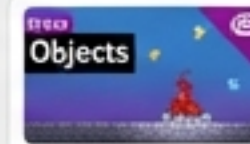
VOCABULARY: SPRITE

A 2D image representing an object (like a player, enemy, or coin).

New object

Asset Store

New object from scratch



Intro: Object types

Watch tutorial

GENERAL



Sprite

Animated object which can be used for most elements of a game.



Tiled Sprite

Displays an image repeated over an area.



Panel Sprite ("9-patch")

An image with edges and corners that are stretched separately from the full image.



3D Box

A box with images for each face

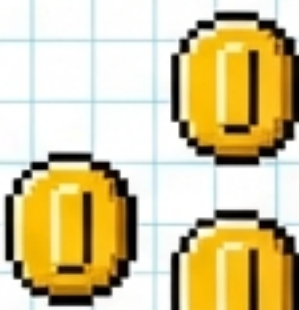


3D Model

An animated 3D model.

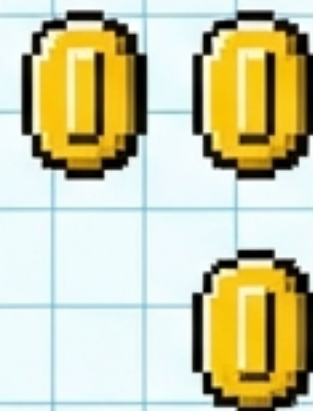


Mission Step: Click 'Add a new object' and select 'Sprite' to begin creating your hero.

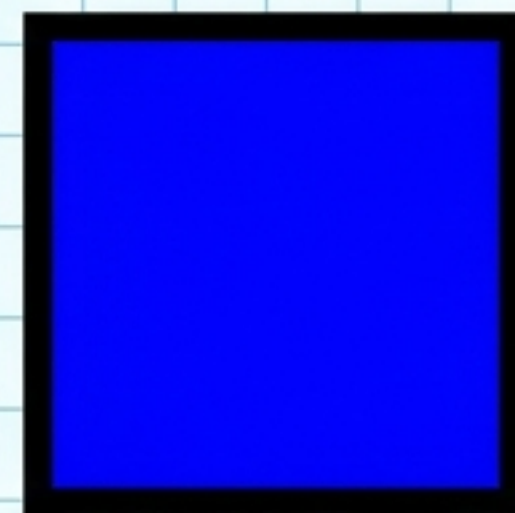
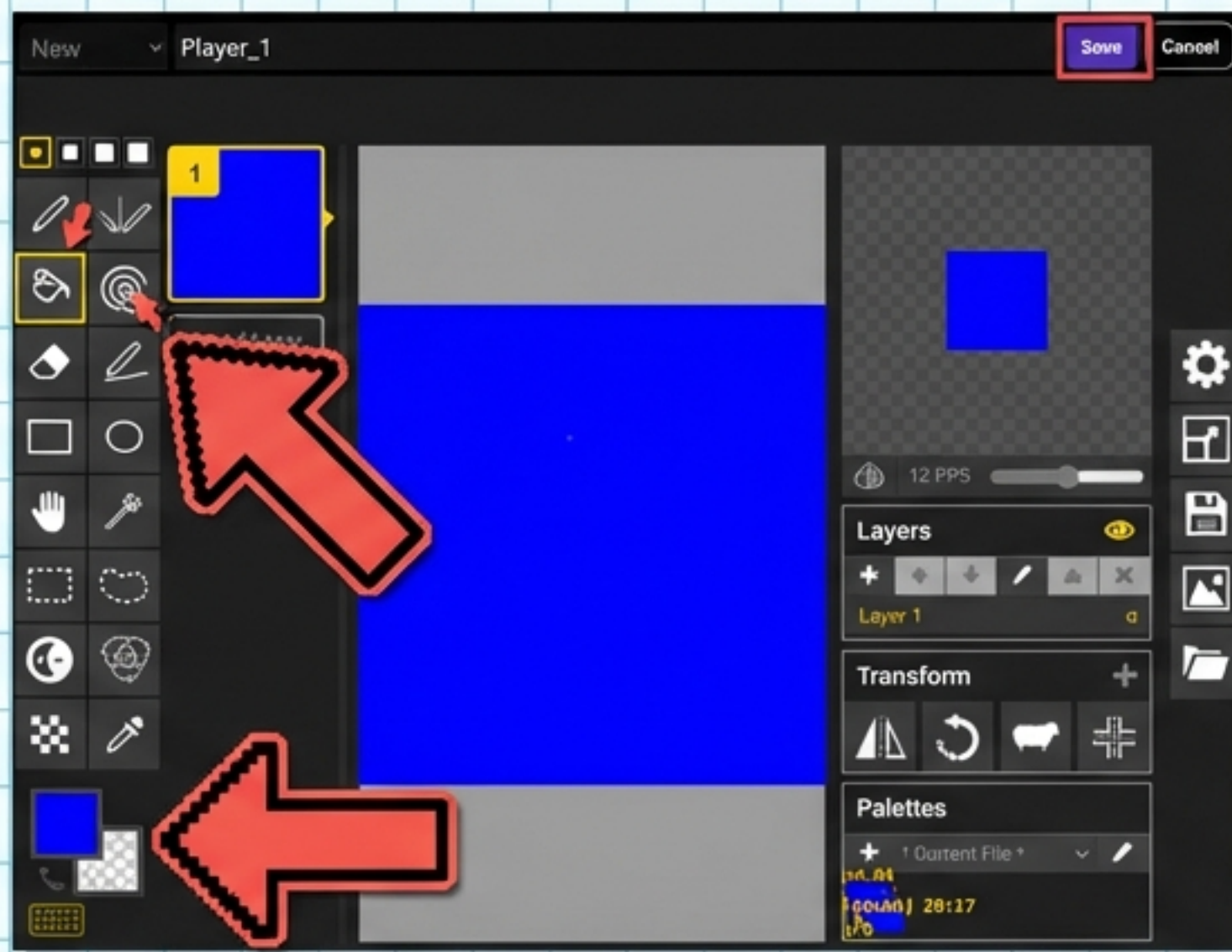




DESIGNING YOUR CHARACTER



1. Select "Create with Piskel".
2. Pick a color (e.g., Blue).
3. Use the Paint Bucket tool to fill the square.
4. Click SAVE.



PLAYER_1





Press Start 2P

ENTER THE PLAYER

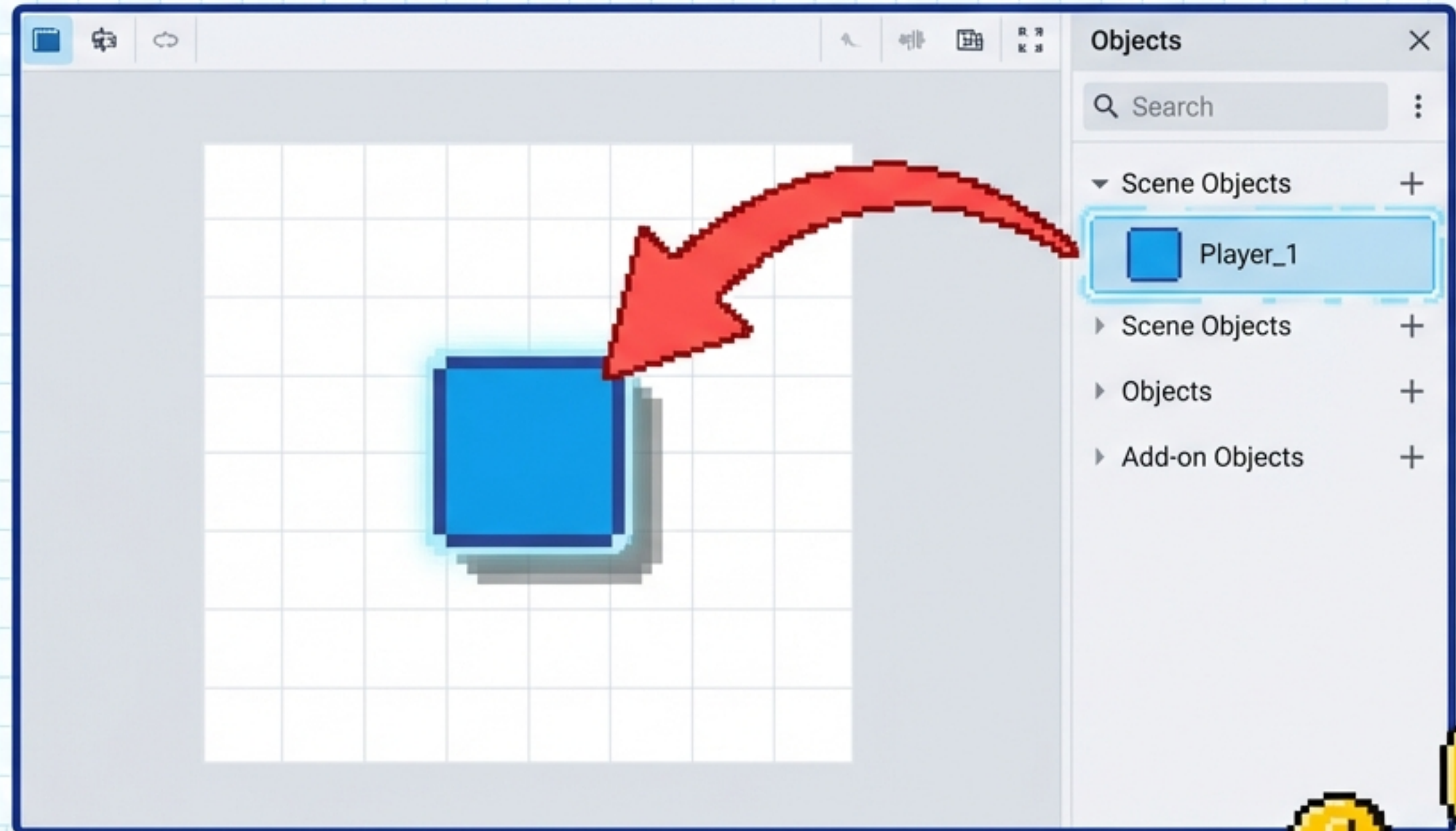


Action:

1. Name your object
"Player_1".

2. Drag and Drop it
onto the Scene.

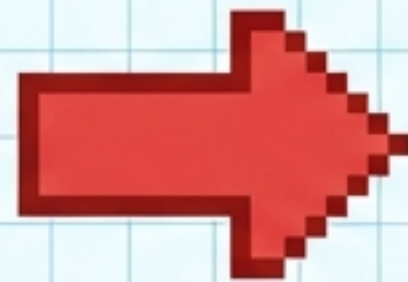
Now your actor is on
the stage!



THE LOGIC: HOW GAMES THINK

EVENT
(The Condition)

**IF I press the
Up Key...**



ACTION
(The Result)

**...THEN the player
moves up!**

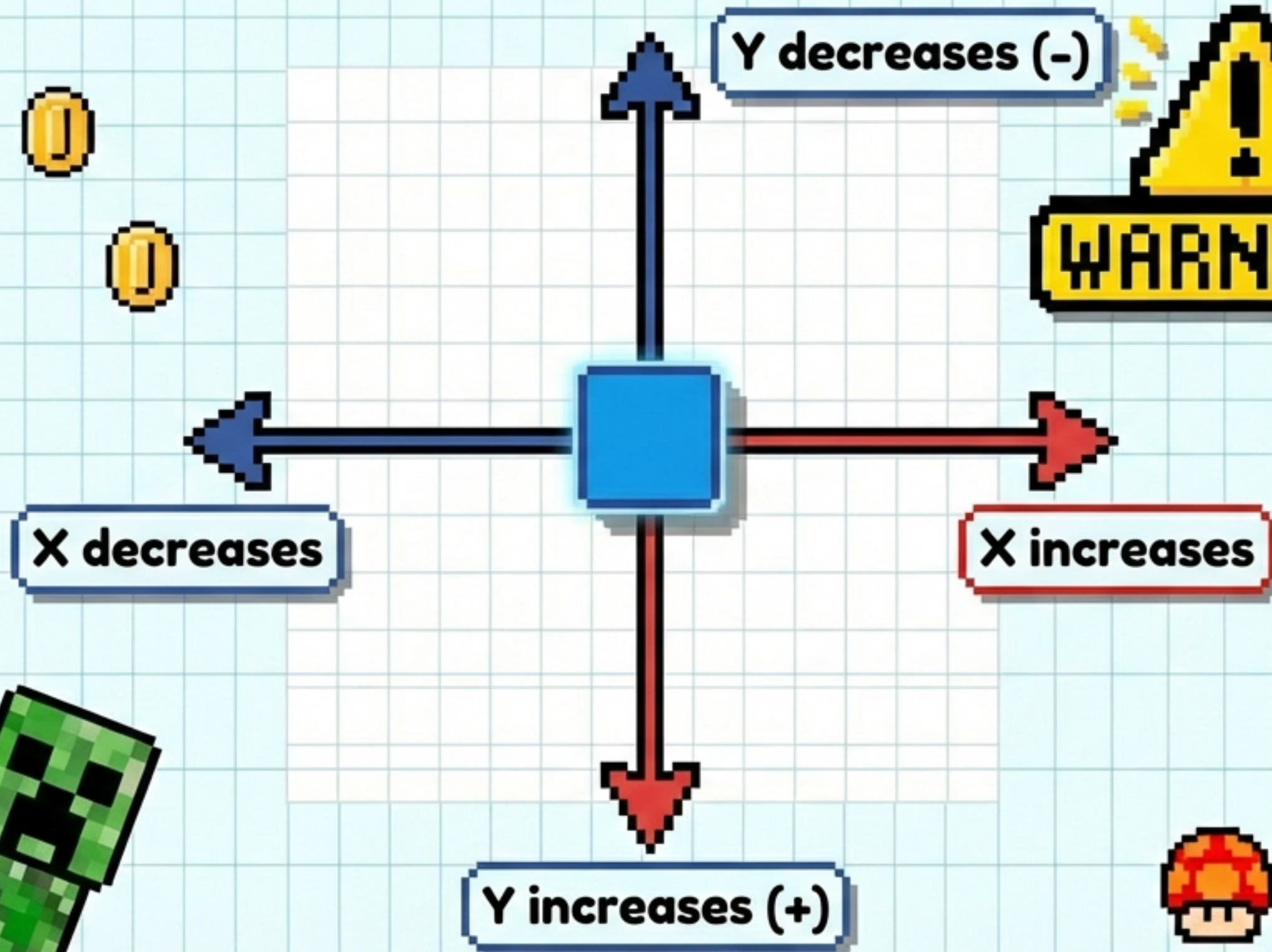
**ITEM ACQUIRED:
EVENT SHEET**

The place where we write the rules of the
game.

+ Add an event

Press Start 2P

THE MATH OF MOVEMENT (X & Y)



CRITICAL ALERT:

In GDevelop:

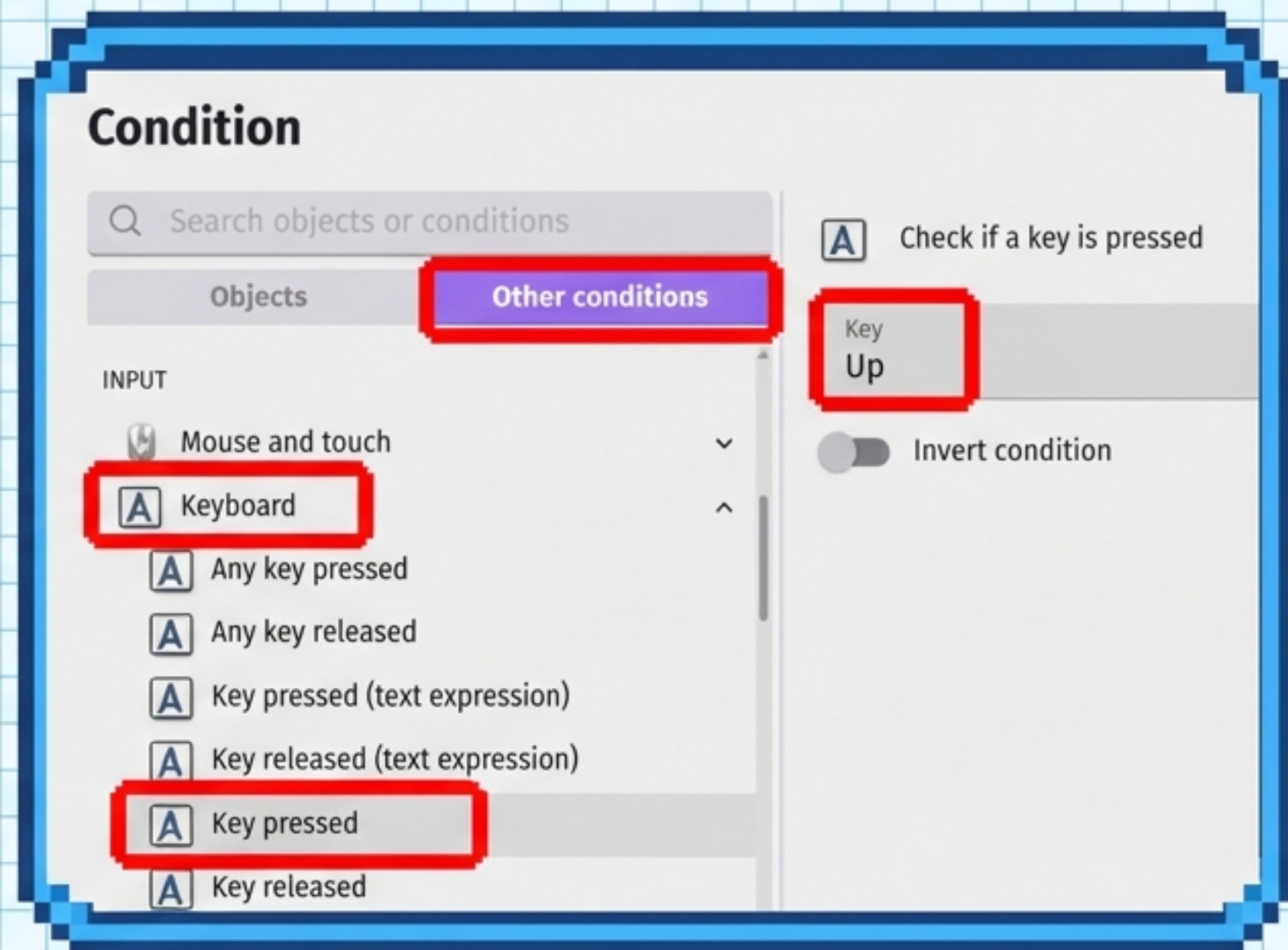
**Subtract (-) from Y
to go UP.**

Add (+) to Y to go DOWN.

STEP 1: THE TRIGGER (CONDITION)

Action:

1. Go to 'Level 1 (Events)' tab.
2. Click 'Add condition'.
3. Search for 'Key pressed'.
4. Select the UP key.



Press Start 2P

STEP 2: THE MOVEMENT (ACTION)



The image shows two side-by-side screenshots of the GDevelop software interface. The left screenshot displays the 'Objects' panel with 'Player_1' selected. A red callout box labeled 'Object: Player_1' points to the object in the list. Another red callout box labeled 'Sign: Subtract (Go Up)' points to the 'Y position' property under the 'Position' category. A third red callout box labeled 'Value: 10 (Speed)' points to the 'Value' input field set to '10'. The right screenshot shows the 'Level 1 (Events)' panel with an event '1.' containing the action 'Change the Y position of Player_1'. The action's configuration shows 'Change the Y position of Player_1: subtract 10'.

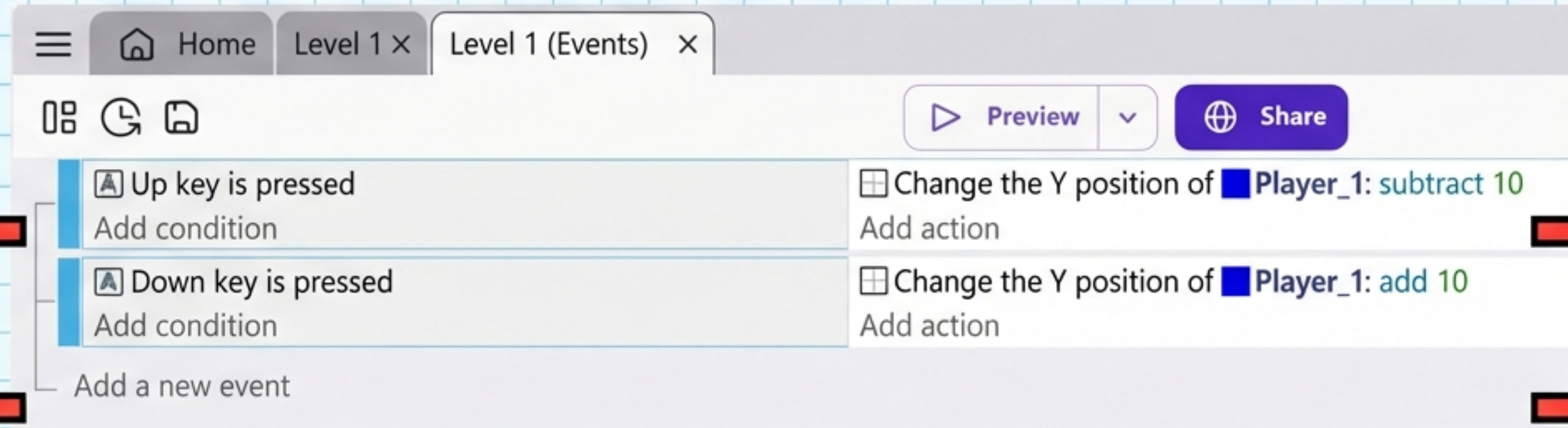
Object: Player_1

Sign: Subtract (Go Up)

Value: 10 (Speed)

Click 'Add Action' to tell the computer what to do when the key is pressed.

CHALLENGE: MOVING DOWN



Home Level 1 × Level 1 (Events) ×

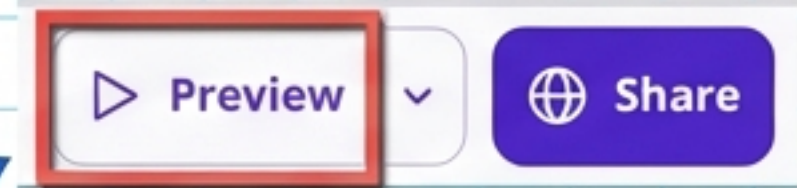
Preview Share

Up key is pressed Add condition	Change the Y position of Player_1 : subtract 10 Add action
Down key is pressed Add condition	Change the Y position of Player_1 : add 10 Add action
Add a new event	

Logic Puzzle: To move DOWN, we must reverse the math.
Condition: Down key is pressed.
Action: Add 10 to Y position.

Press Start 2P

MISSION SUCCESS: TEST YOUR GAME



1. Click the **PREVIEW** button.
2. Press your Up and Down arrows.
3. Does it move?

CONGRATULATIONS!
You are now a Game Designer.

CHECKPOINT CHALLENGE

QUEST 1

_____ are visual representations of an object.

QUEST 2

Rules are written in the _____ panel.

QUEST 3

The _____ button is used to test the game.



Answer Key
1. Sprites
2. Event Sheet
3. Preview

Press Start 2P

MISSION INVENTORY (RECAP)



Sprite:
The 2D image.



Scene:
The level design area.



Event:
The rule
(If/Then).



Object:
The active
game element.

NEXT MISSION: Adding Left and Right Horizontal Movement!

